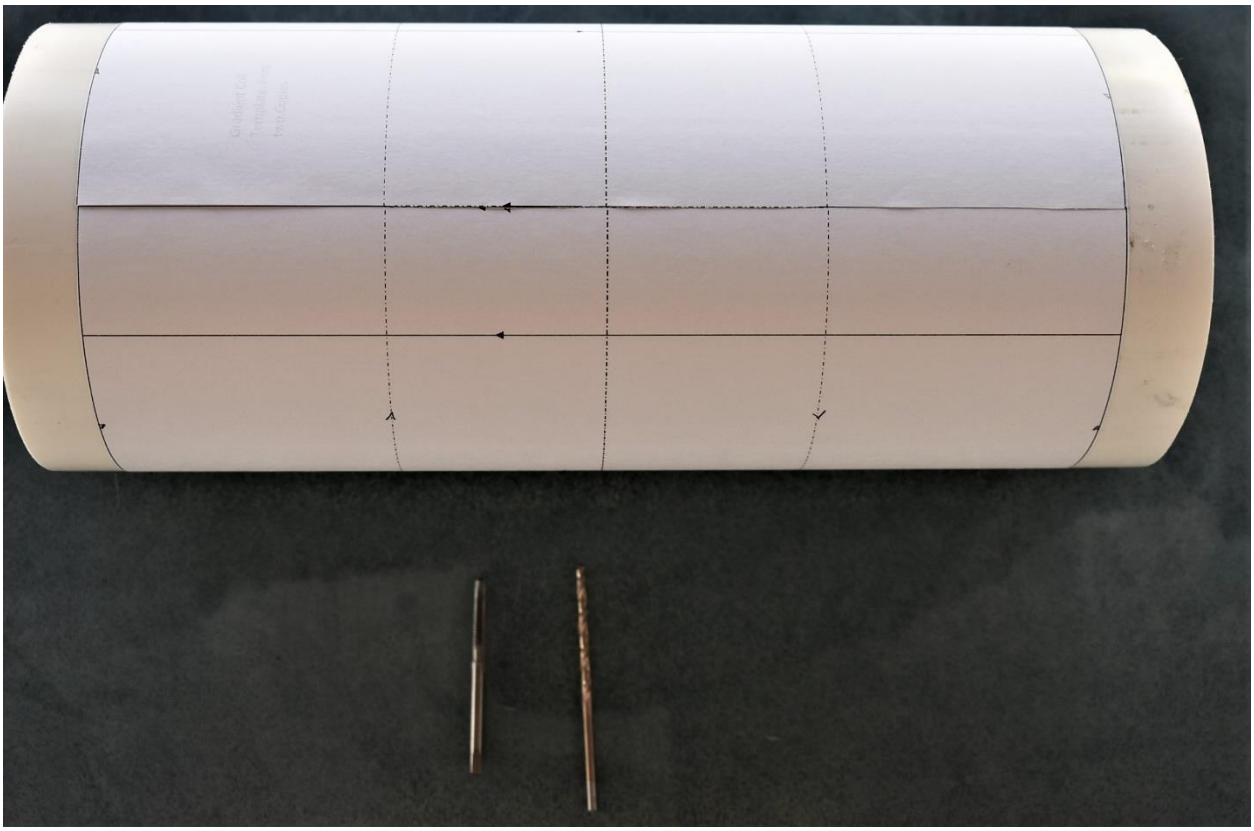
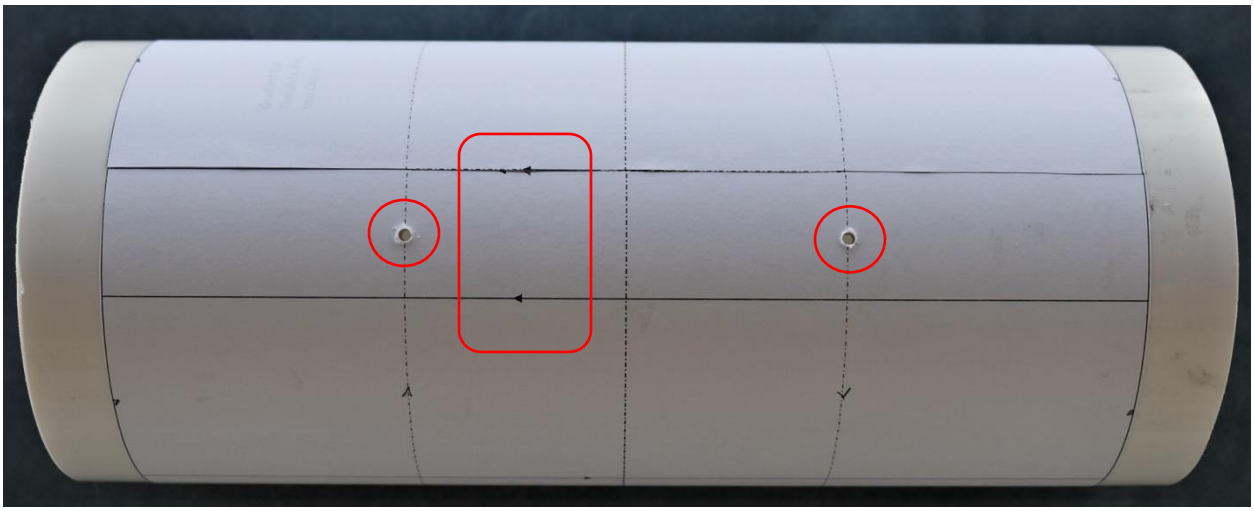
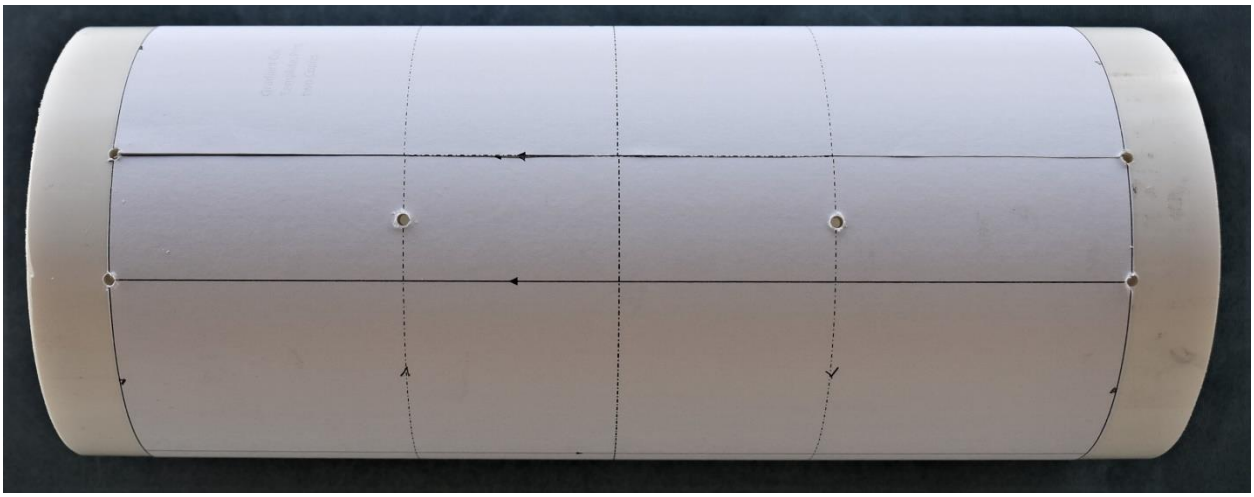
Coil Winding, PCB parts layouts, and harness connections



Glue the two template sheets together. Wrap the sheet around the 4” PVC pipe and glue the sheets at the overlap. You will need a 7/64” bit, a 6-32 tap and an electric drill.

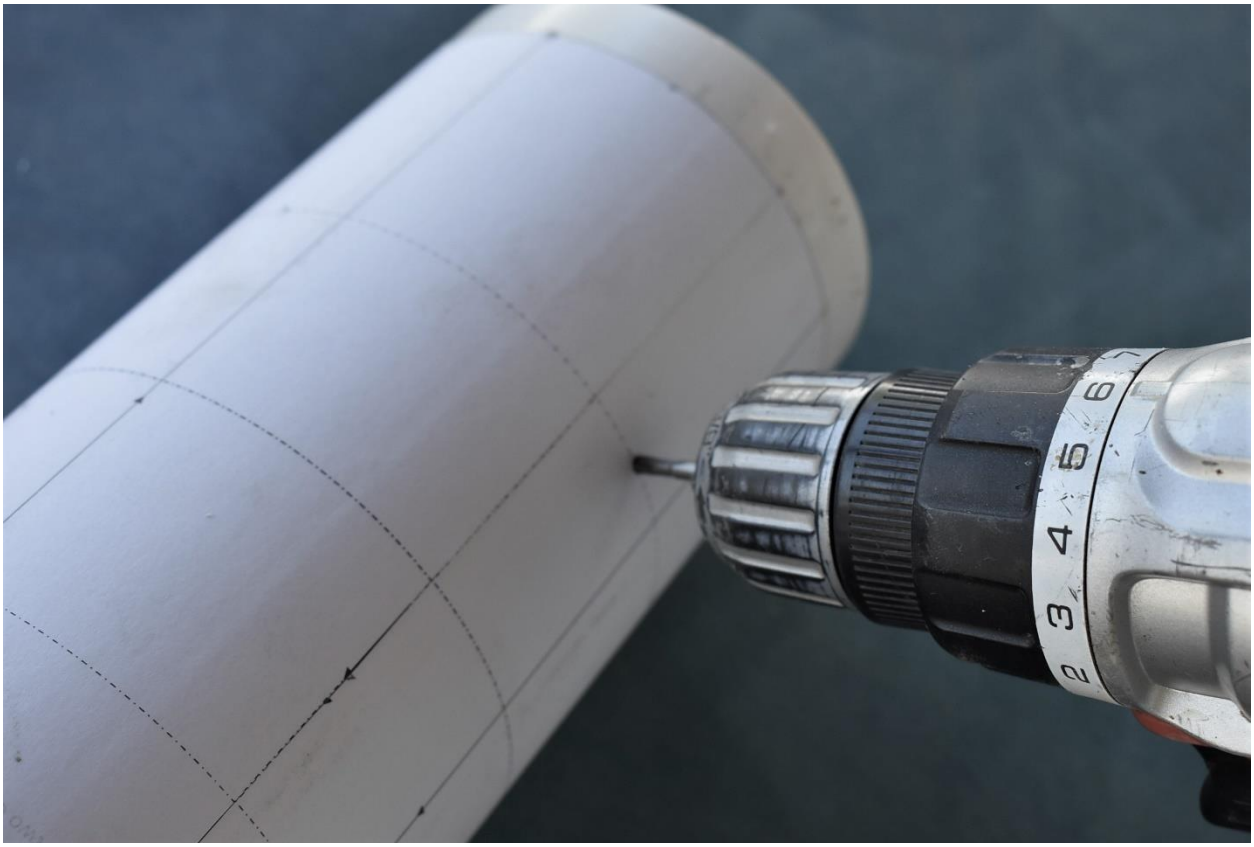


First, drill two holes on the Helmholtz coil lines. The holes are centered between the quadrupole coil lines that point in the same direction to the left.



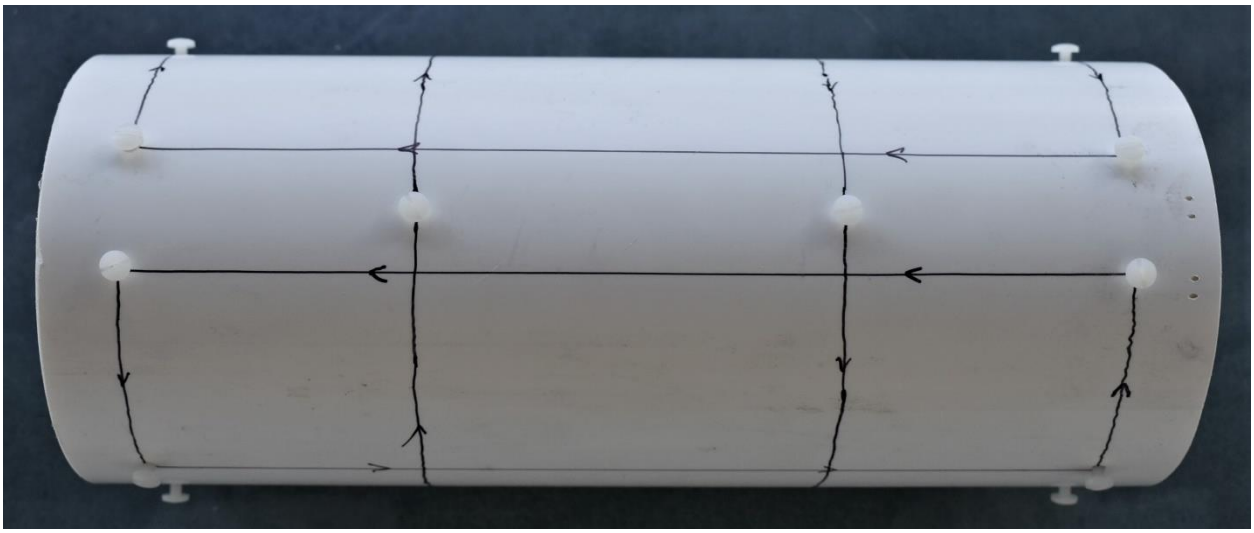
At the corners of the quadrupole coil lines, drill a hole. There are four coils. So, 16 holes.

At the edge of the 4” PVC pipe, drill two pairs of holes angled to the outer diameter. The bit is 3/64”. The pair of holes with the bit is aligned with the two holes on the Helmholtz coil lines.

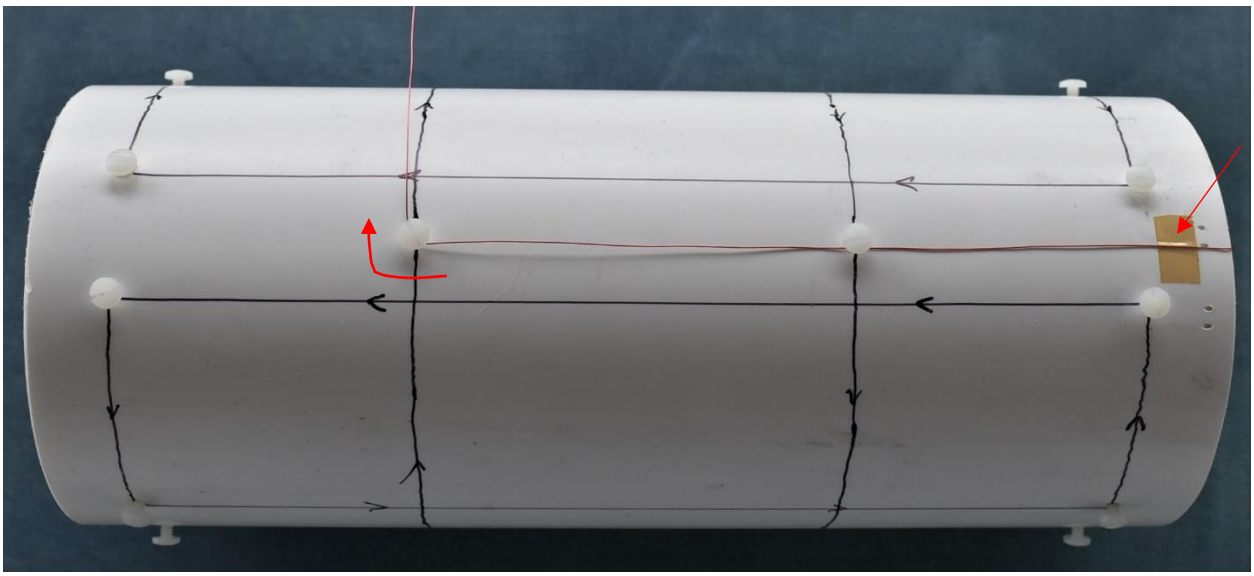


Use the drill to run the tap in and out of the drilled holes.

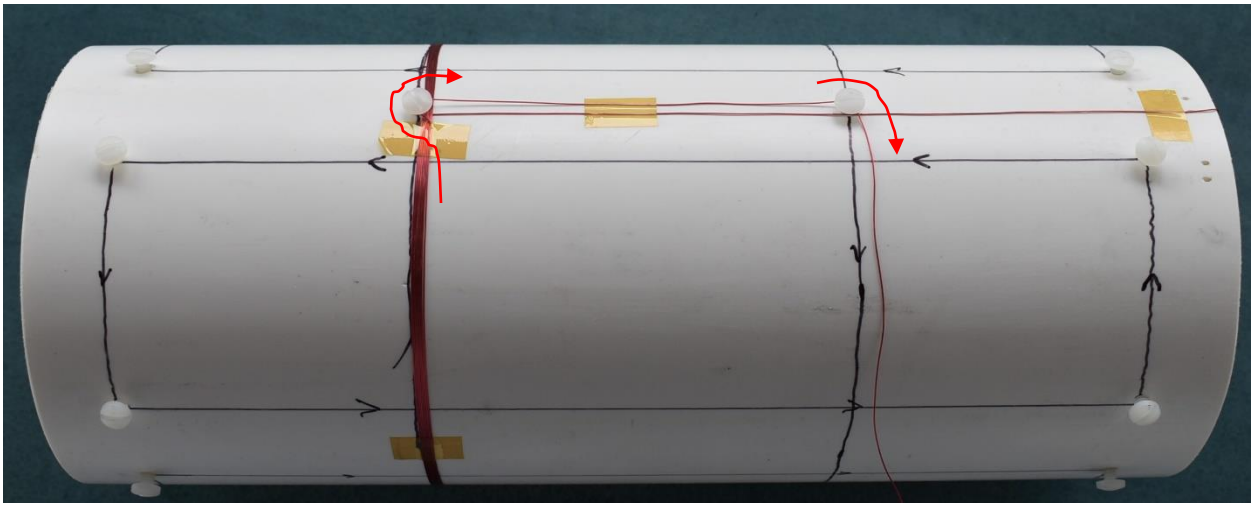
Remove the paper template. Use a felt tip pen to replicate the winding patterns on the paper template.



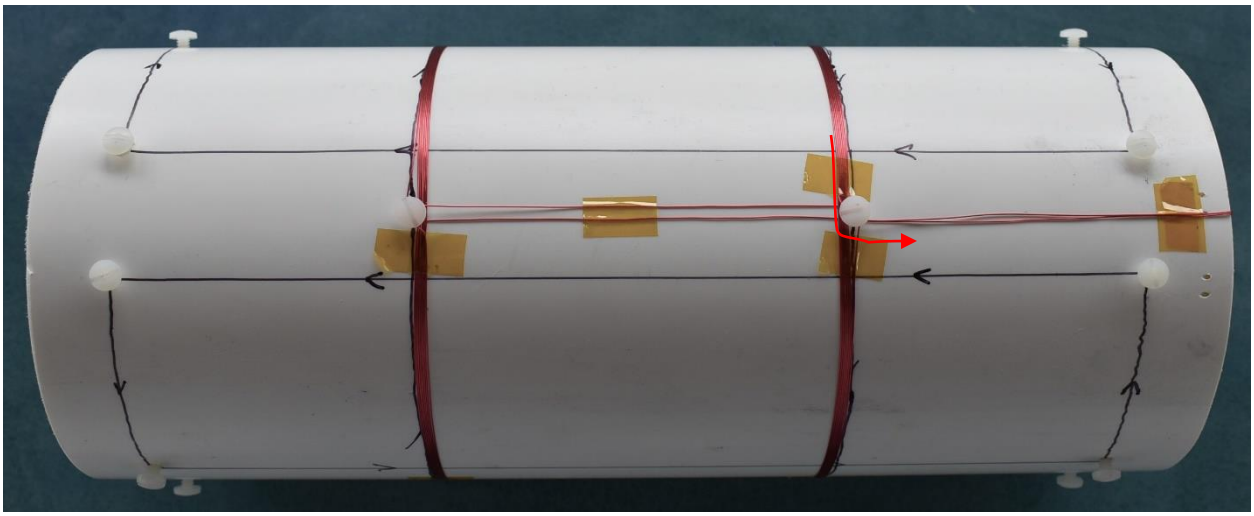
Obtain 18 nylon screws 3/8” long 6-32 threads. Thread the screws in until they are flush with the internal diameter of the pipe.



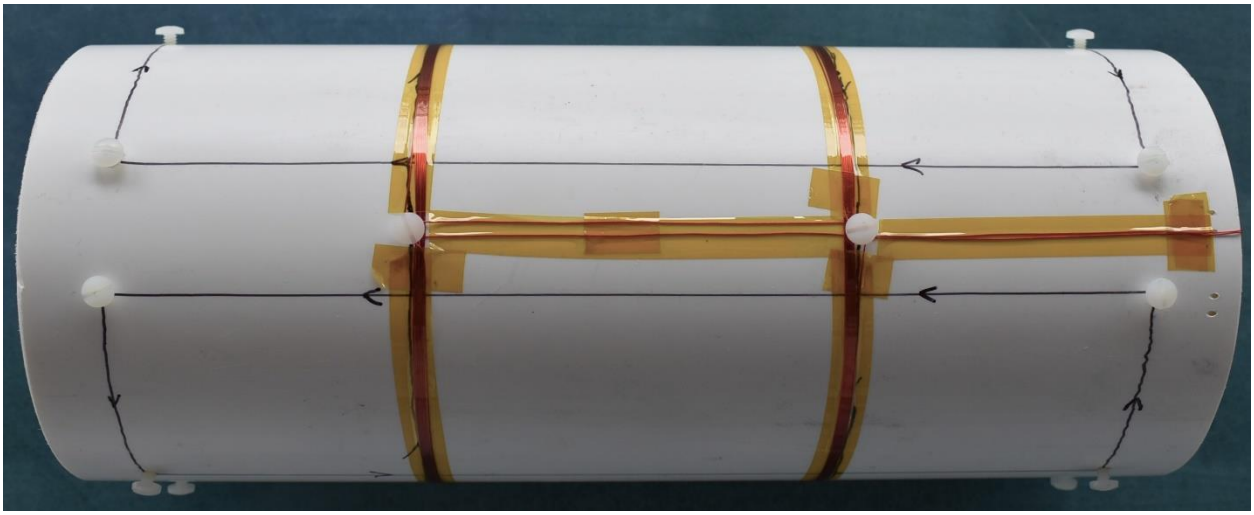
Tape a 28 gage insulated magnet wire to the PVC pipe. Leave tail of several inches and tuck it into the inside diameter of the pipe. Wrap the wire counter clockwise around the left Helmholtz coil screw.



Wrap 8-turns in the up direction. The windings are located to the right of the screw. Place a small piece of tape to hold the windings. Wrap the wire clockwise around the left screw then clockwise around the right screw.

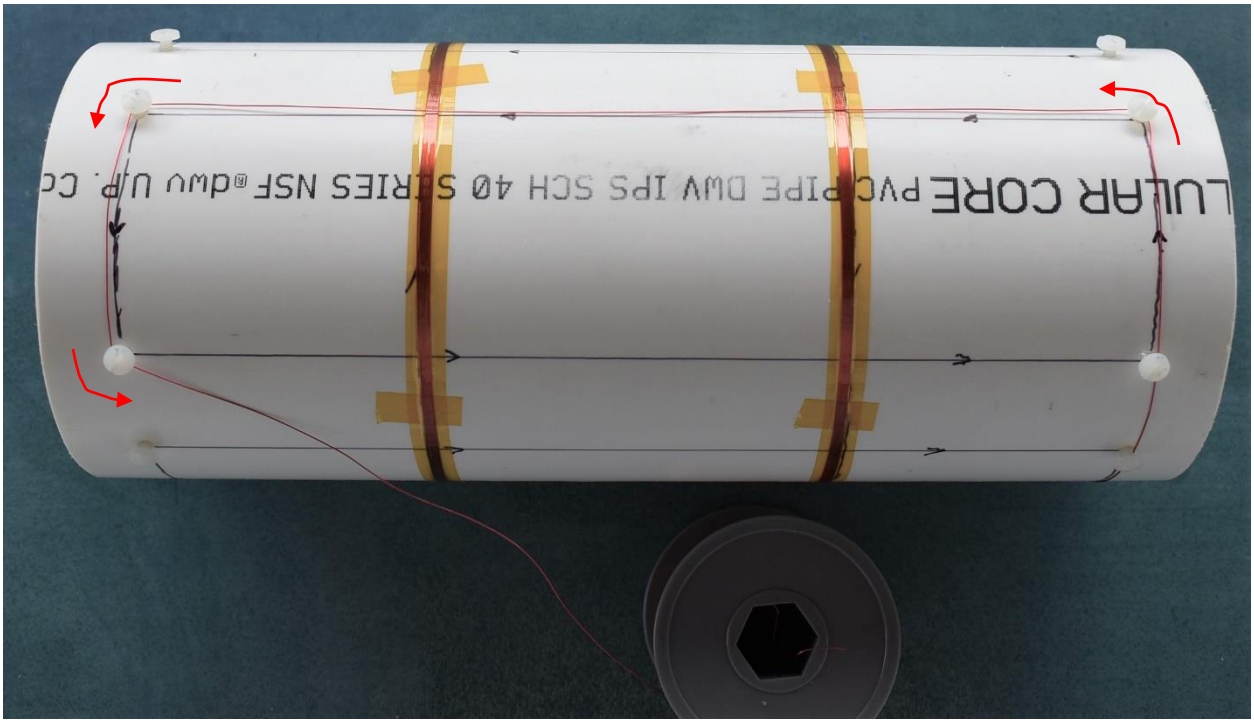


Wrap 8-turns in the down direction. The windings are located to the left of the screw. Place a small piece of tape to hold the windings. Wrap the wire counterclockwise around the left screw. Tape the wire at the edge of the pipe. Cut the wire, leaving several inches and tuck them into the pipe.

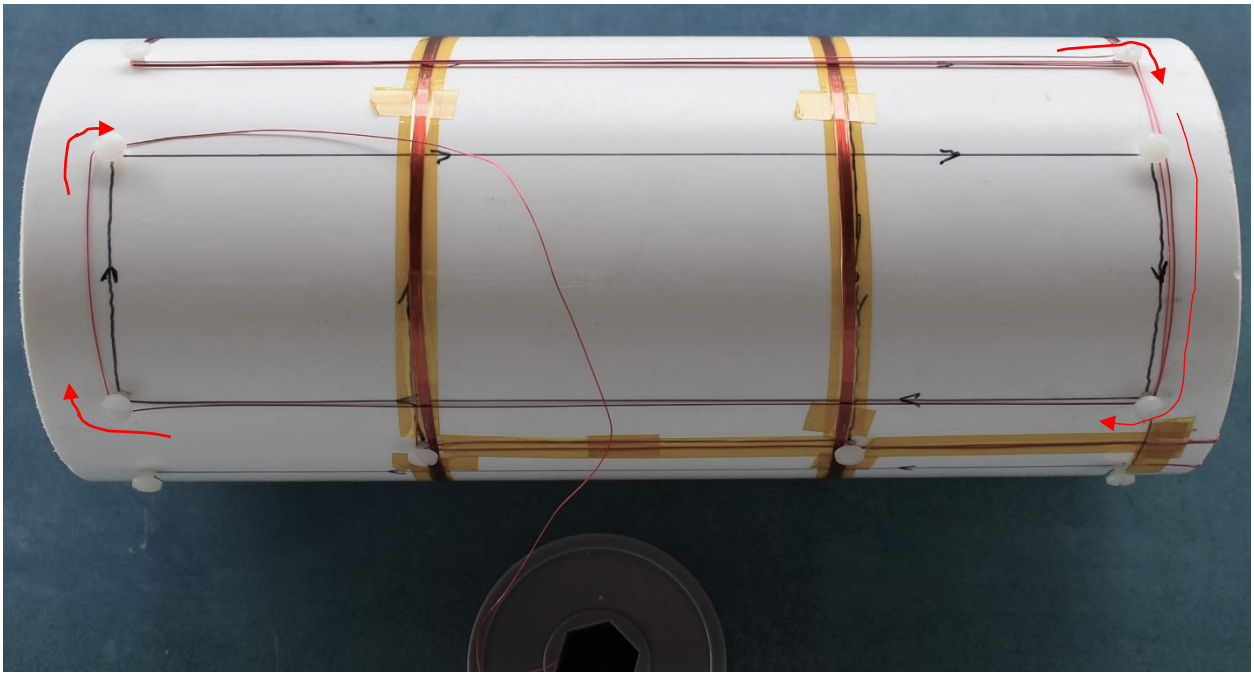


Place tape over the windings and the feed path.

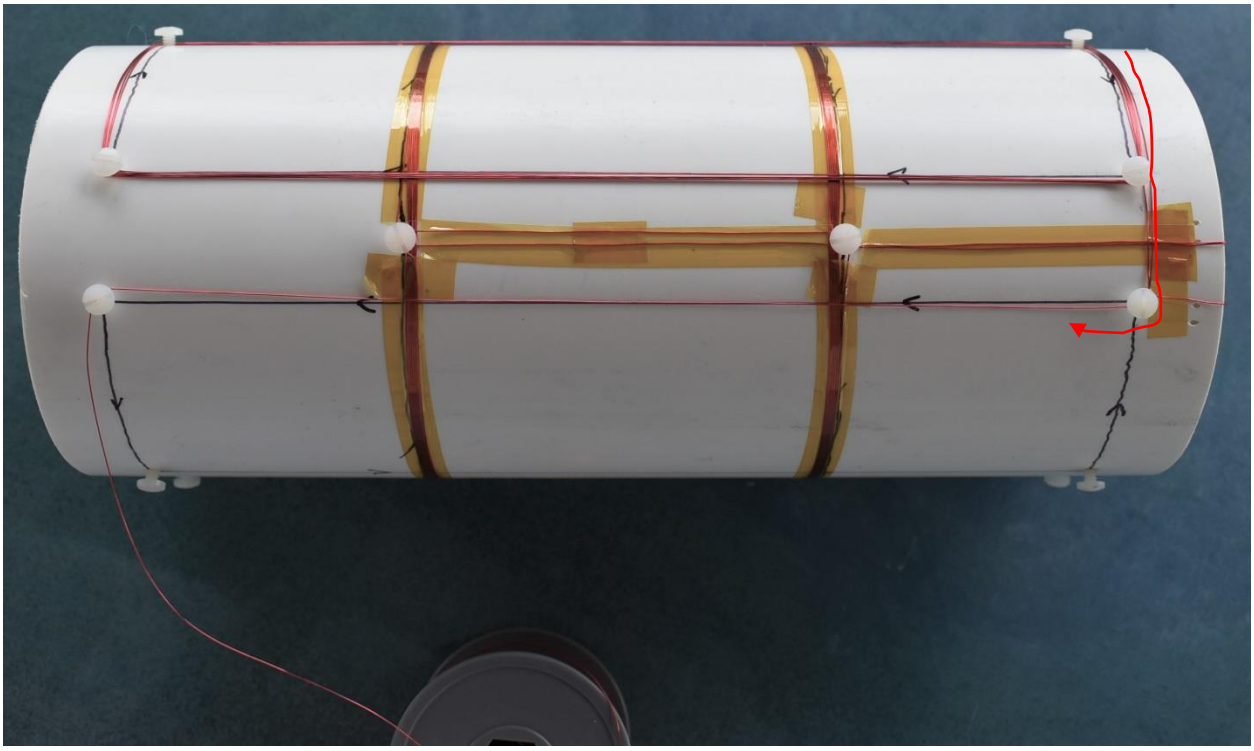
Tape a piece of magnet wire to the left edge of the pipe near the second set of angled holes. Bend the wire up wrapping it around the pipe. It does not wrap around the nearby screw.



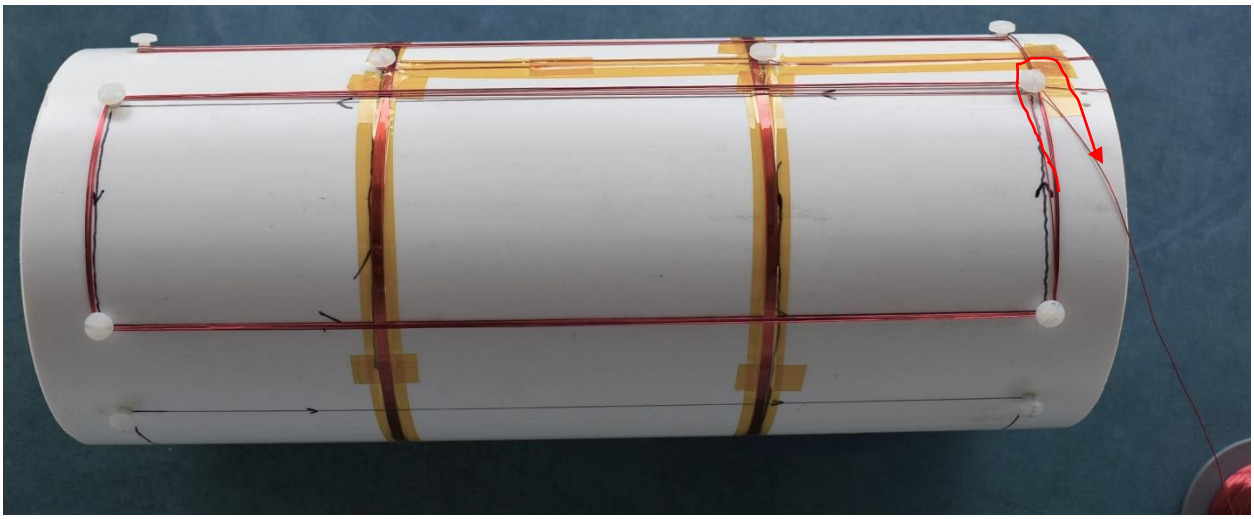
Wrap the wire past the first quadrupole loop to the second loop. The wire loops around the upper right screw and winding is counterclockwise. Wrap four loops around the screws.



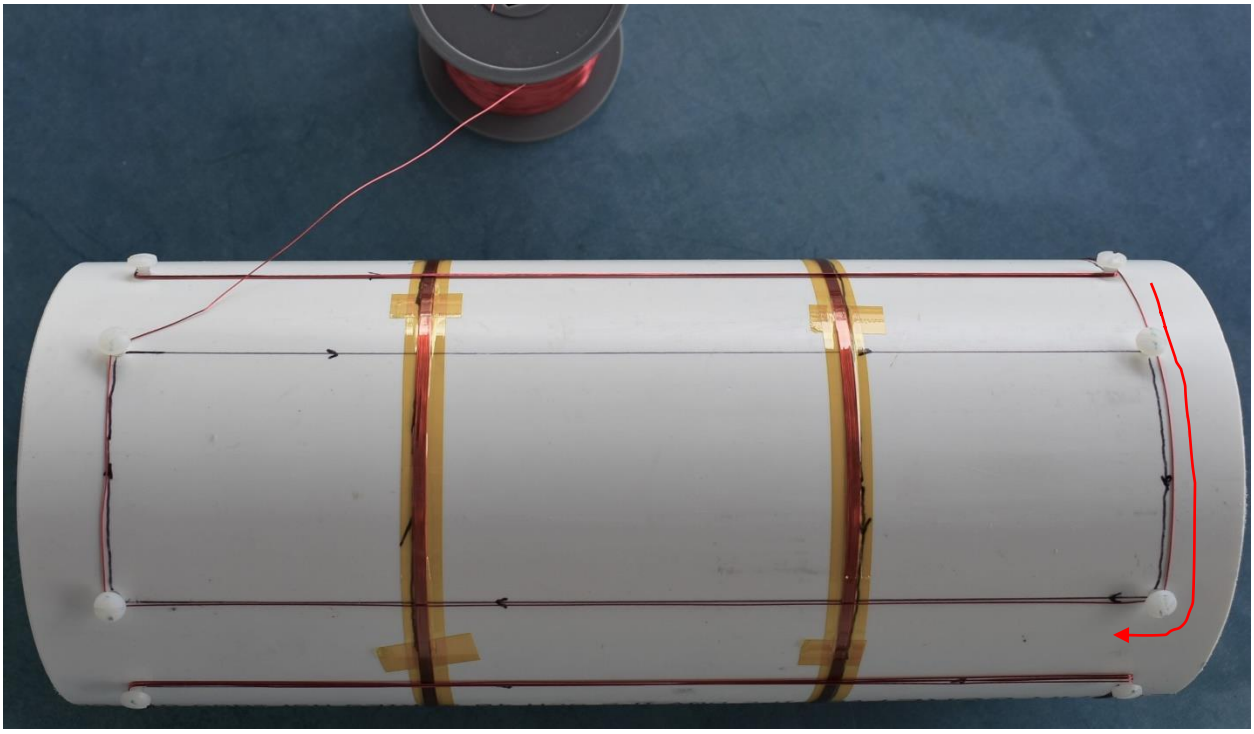
After the forth loop is completed, wrap the wire clockwise over the screw and drop it down to the lower right screw on the next coil. Wrap four clockwise turns on the coil



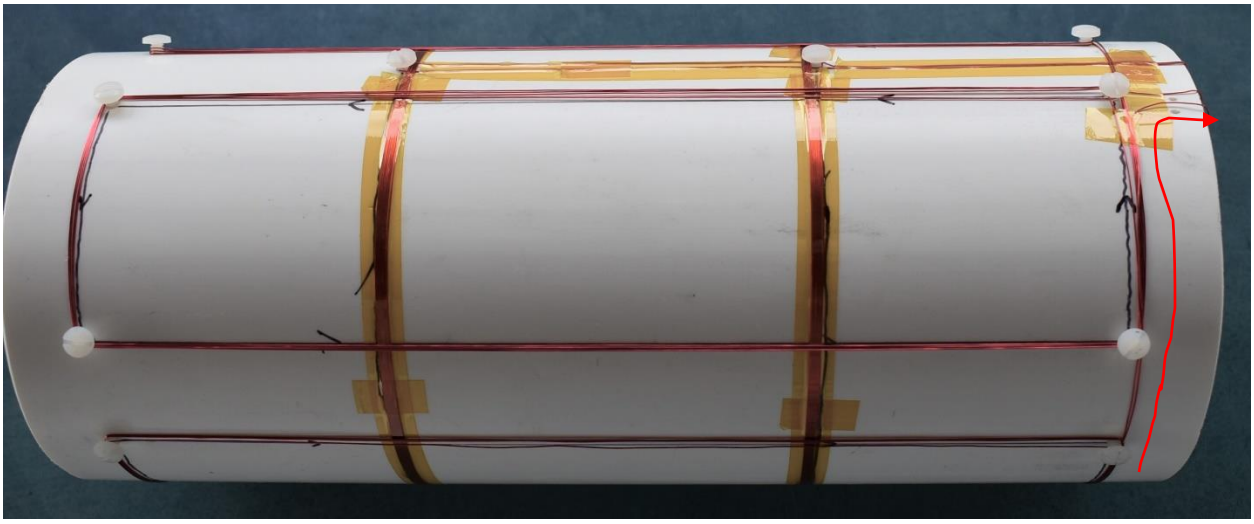
After 4-turns, continue the wire to the next quadrupole coil. Wrap the wire clockwise around the upper right screw and wind in the counterclockwise direction.



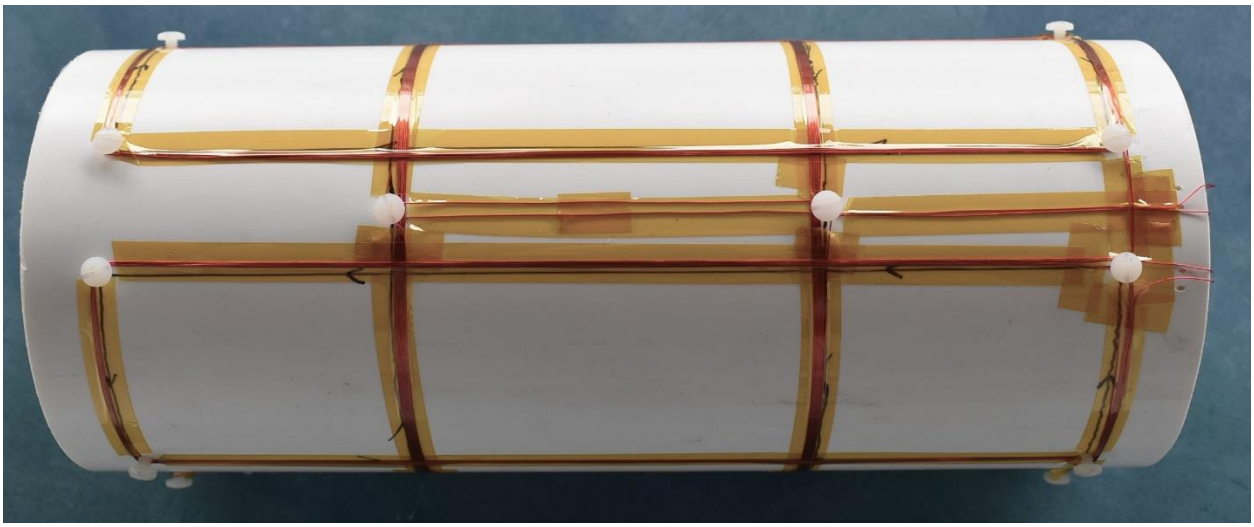
Wind 4-counterclockwise turns then make a clockwise loop around the upper right screw.



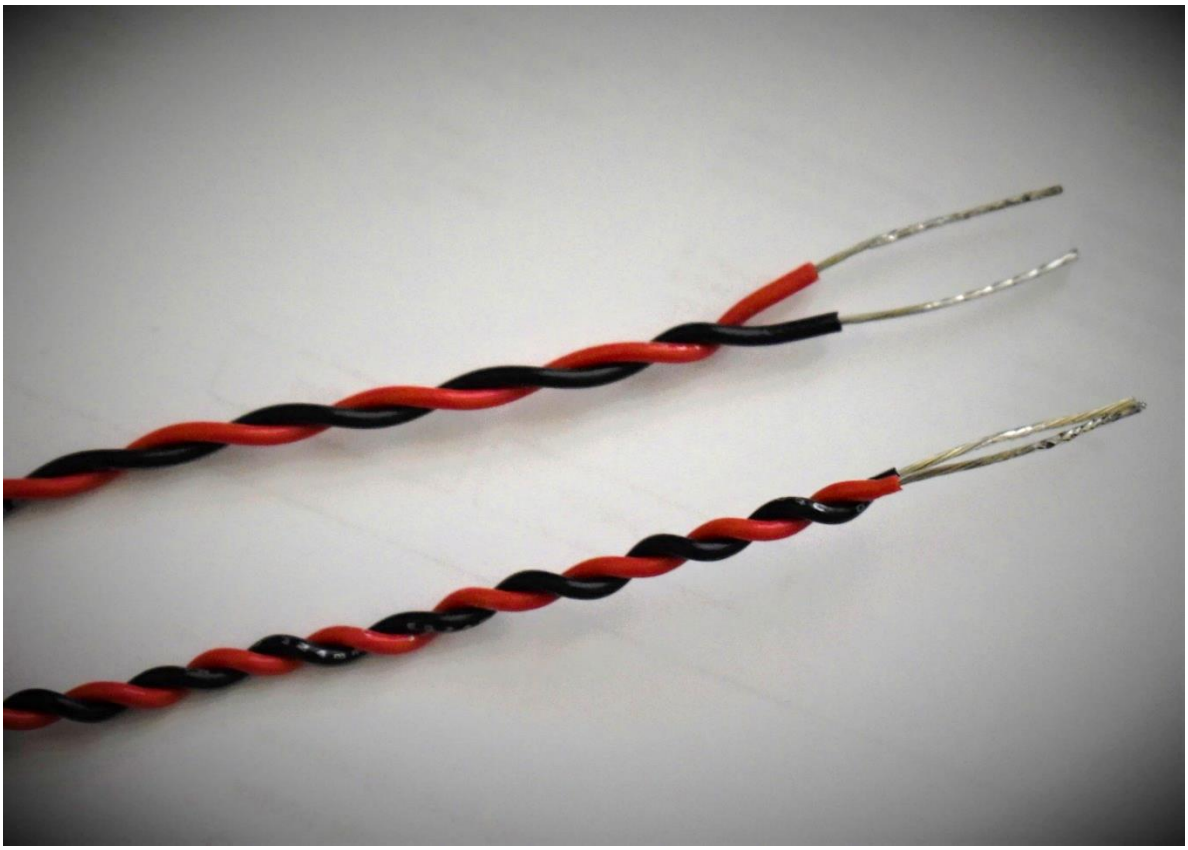
Route the wire counterclockwise around the bottom right screw of the last quadrupole coil. Wind four clockwise turns. Finish at the bottom right screw. Route the wire counterclockwise around the screw and back up.



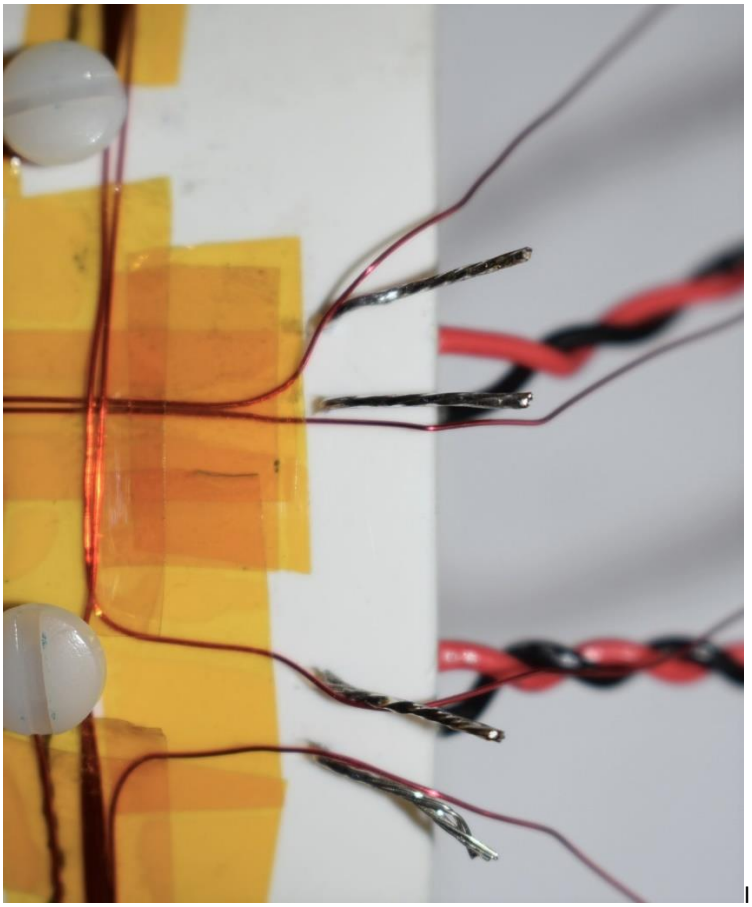
The wire returns to the drilled hole pair where it is taped in place. Leave a few inches of wire. Cut and tuck it in the pipe.

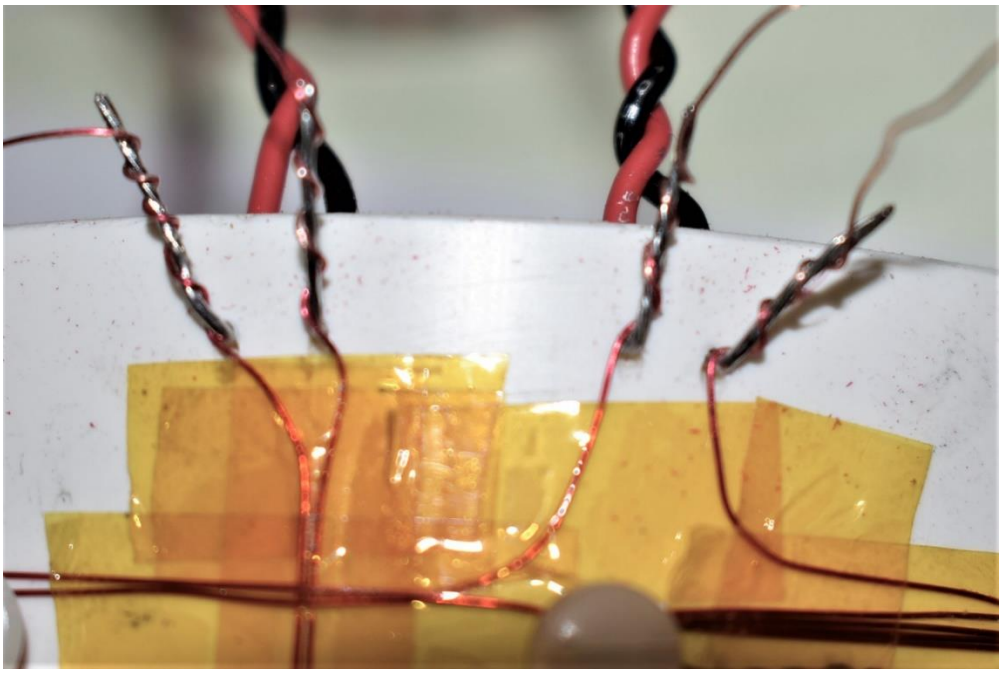


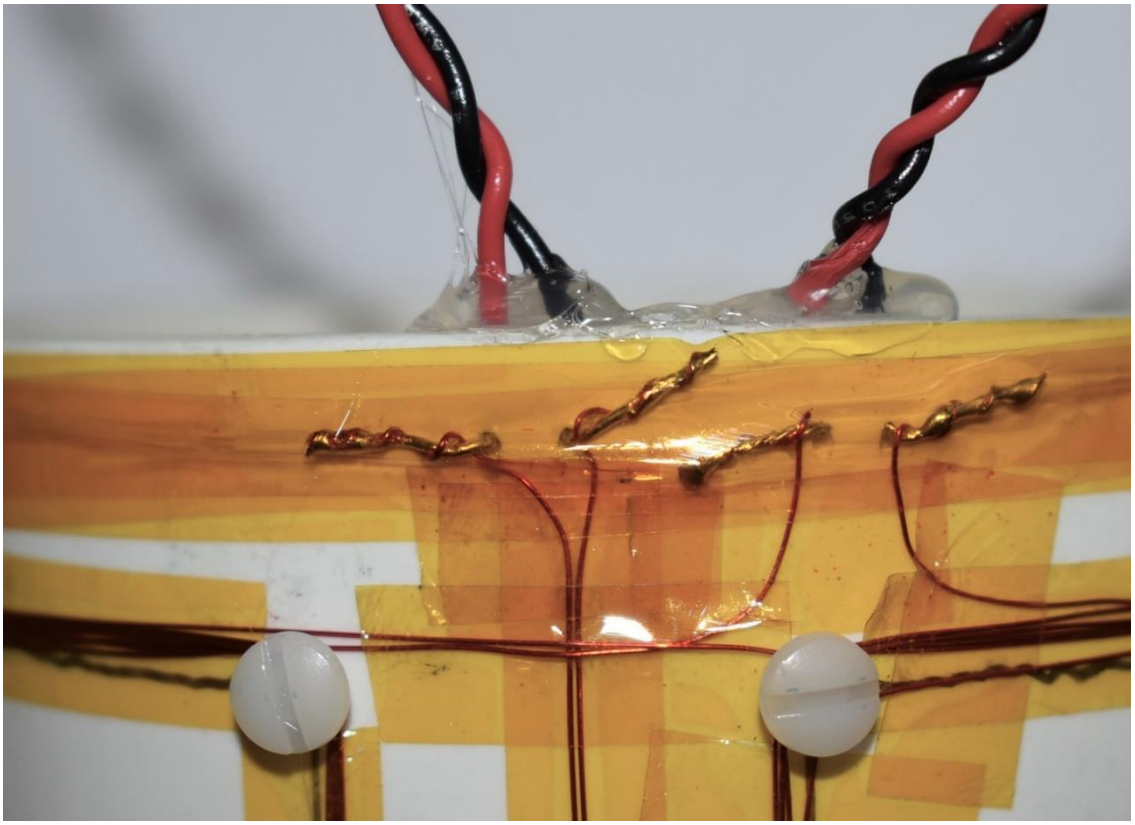
Tape the quadrupole windings to the pipe.



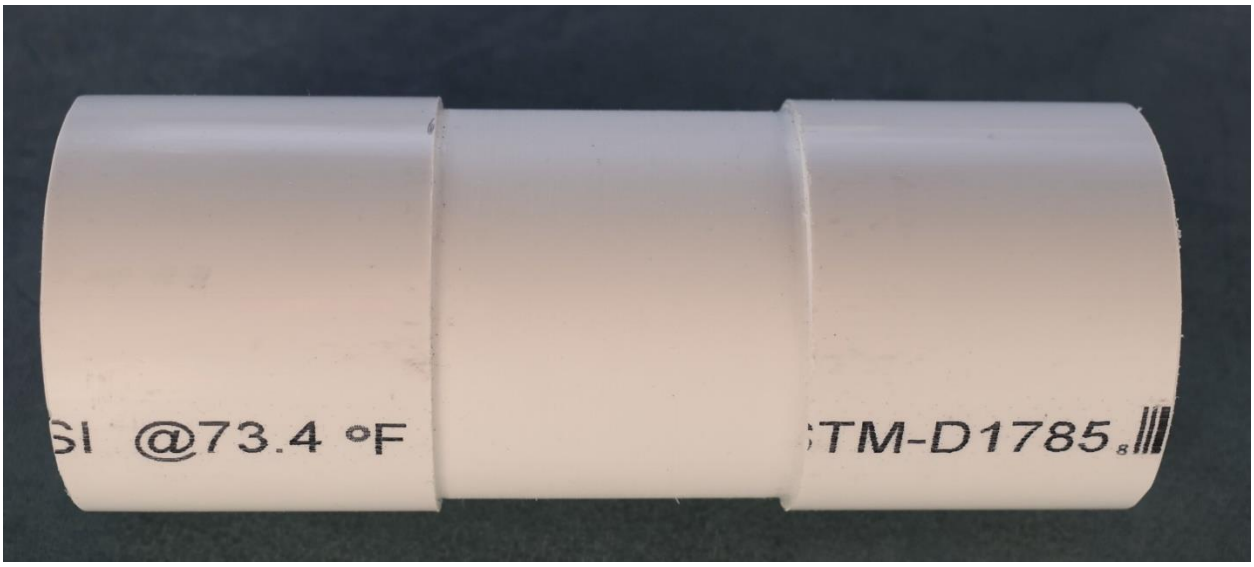
Prepare two 2-foot lengths of black and red 22 gage stranded wire. Twist the pairs together and strip 1” of insulation from each end. Tin the wires with solder.


Insert the wire pairs through the end holes.

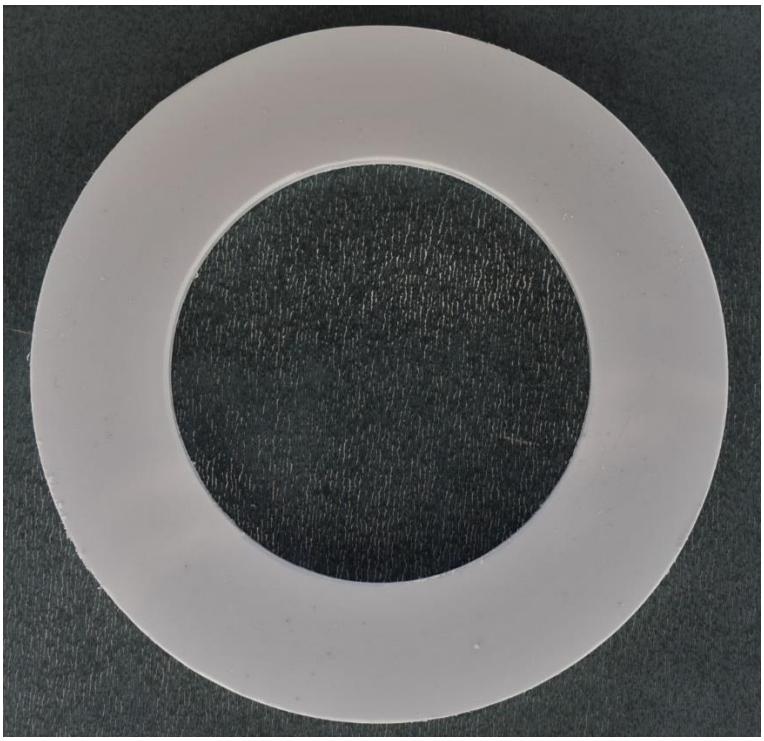

Use sandpaper to remove the varnish. Wrap the magnet wires around the tinned leads. The Helmholtz coil connects to the left pair and the quadrupole coil connects to the right pair.



Solder the wires to the leads. Bend the wires as not to short and tape them in place. Use hot glue to stress relieve the twisted pair feeder wires.

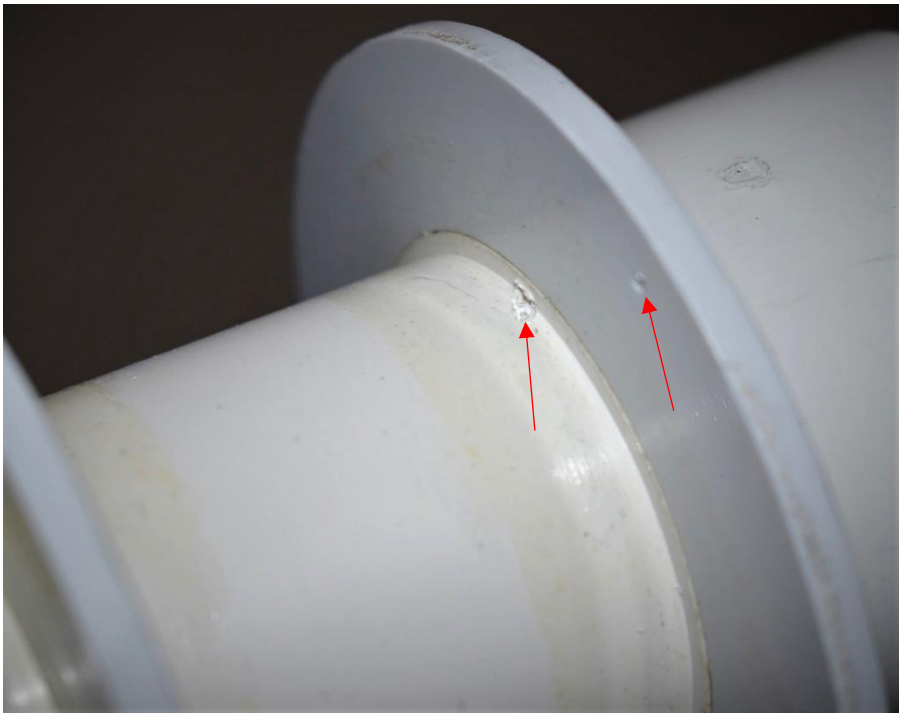


Use a lathe to turn the center of the 2” PVC pipe to an outside diameter of 2.25”. The cut is 2” wide. You will need two pipes.

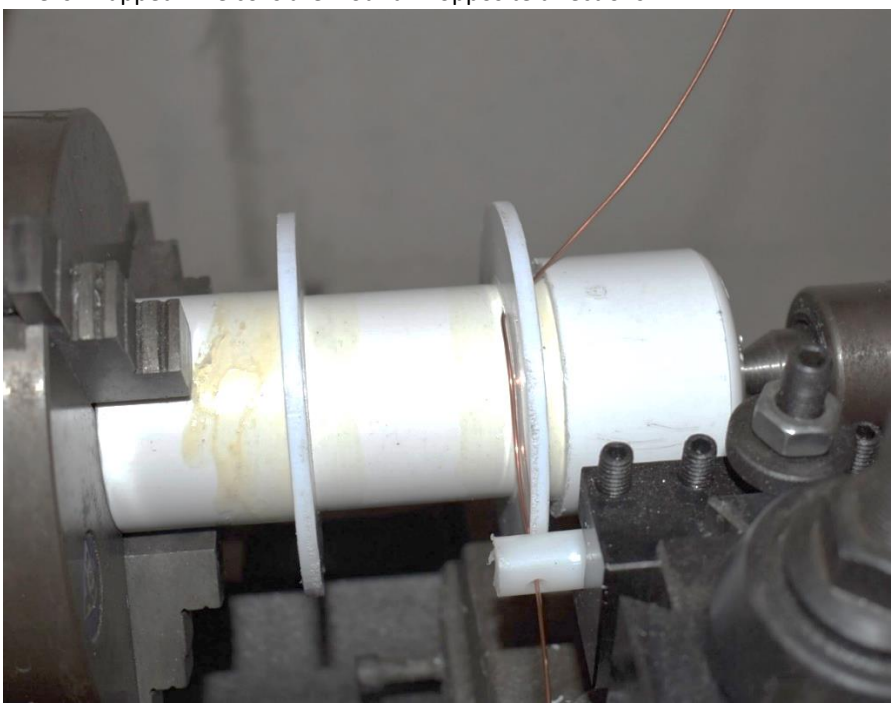


Cut four coil forms. The material is 3/16” PVC sheet. The doughnuts have a 2.4” I.D. and a 3.99” O.D.

Two forms are glued with PVC cement flush with the turned center of the pipe. The forms hold the sidewalls of the coil pack.

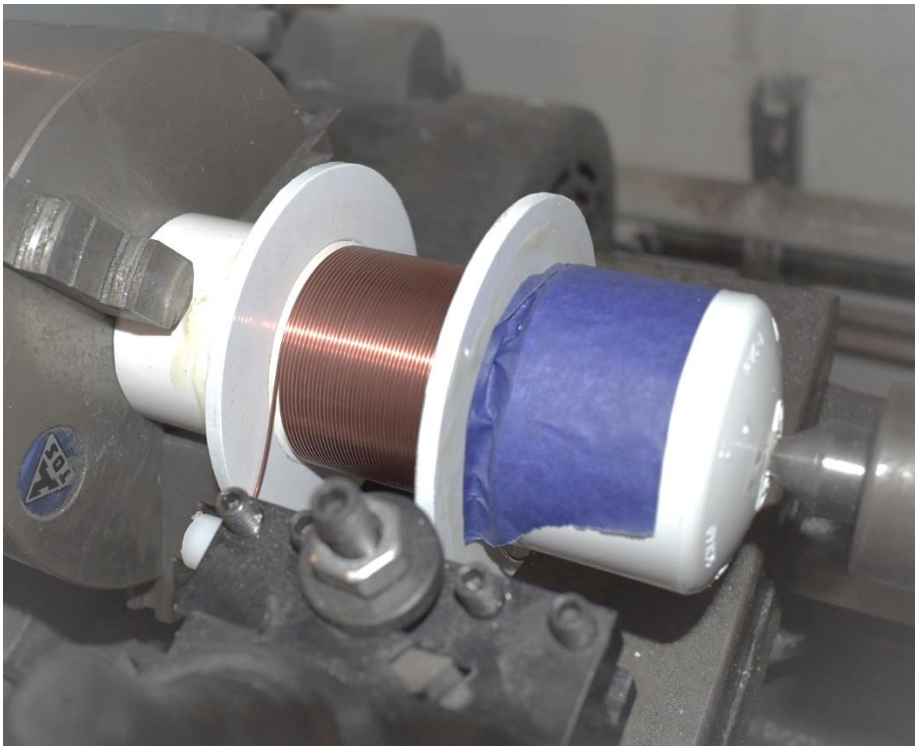


Two 3/64” holes are drilled into one sidewall of the coil form. The top one is straight and it is down about ¼” from the top edge. The other is drilled at an angle. It is flush to the bottom of the form and it exits in the direction the wire is wrapped. The coils are wound in opposite directions.

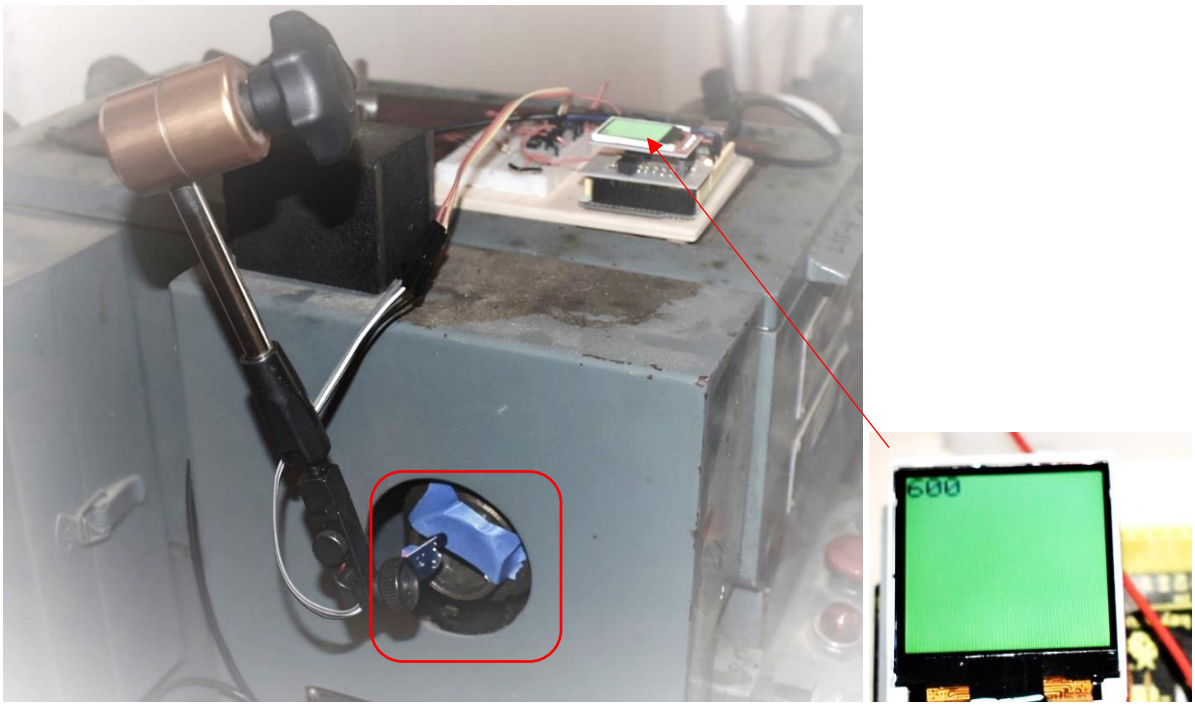


I use a drilled piece of nylon round for a wire guide. The apron moves at 0.046” per turn which is slightly greater than the 0.043” diameter of the wire. This coil was wound with the motor reversed for the CW direction. The next coil was wound with the motor in the normal cutting direction for the CCW direction.

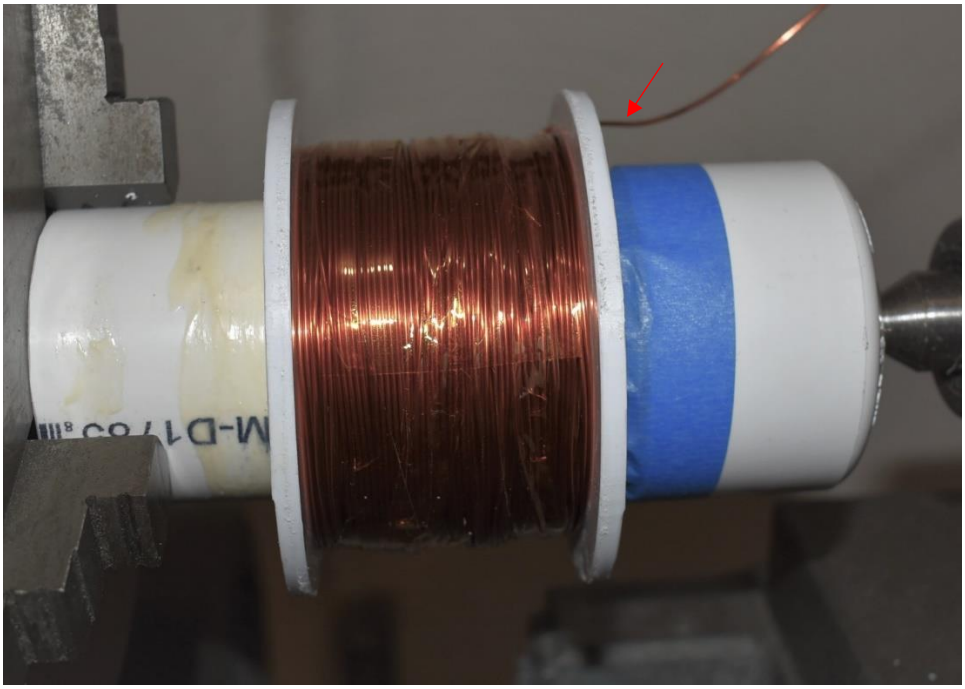
The magnet wire used was 18AWG high temperature (200C) insulated. I used it for the tougher insulation to reduce the possibility of damage during winding.



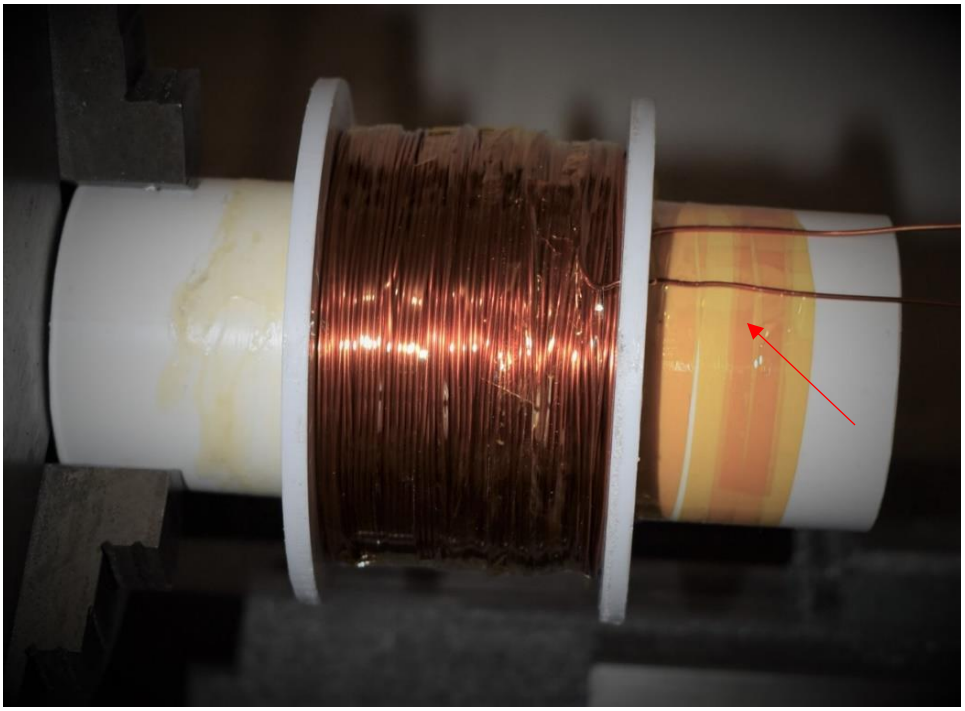
Abount 43 turns are wound per pass. It takes at least 14 passes. The apron direction is reversed at the end of each winding pass. It is possible to get about half of the windings to pack uniformly. After that, you may see some gaps and bunching. This does not have a great effect on the coil. My coils match within 1%.



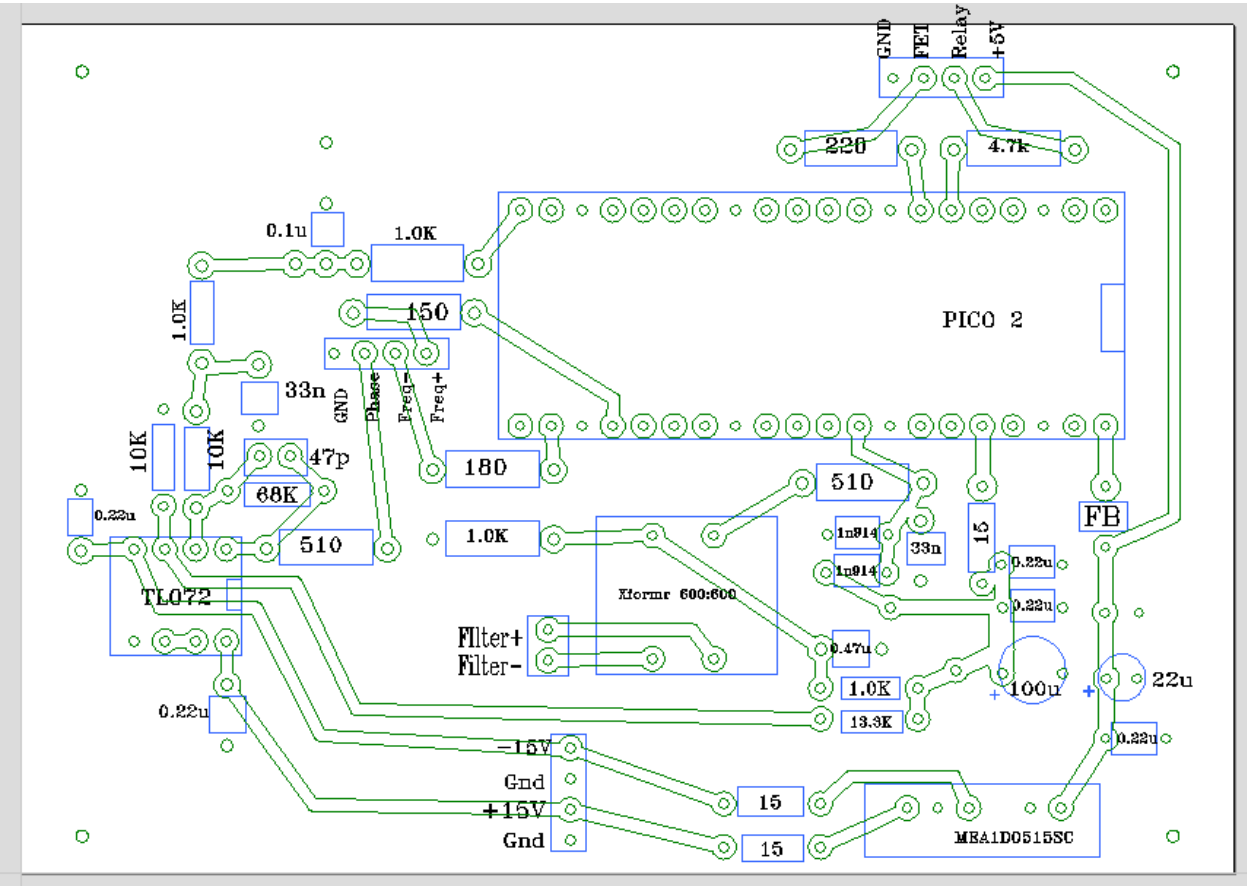
I used an Infrared reflection detector and a piece of tape to make a turns counter. An Arduino kept track of the pulses and displayed a simple count.



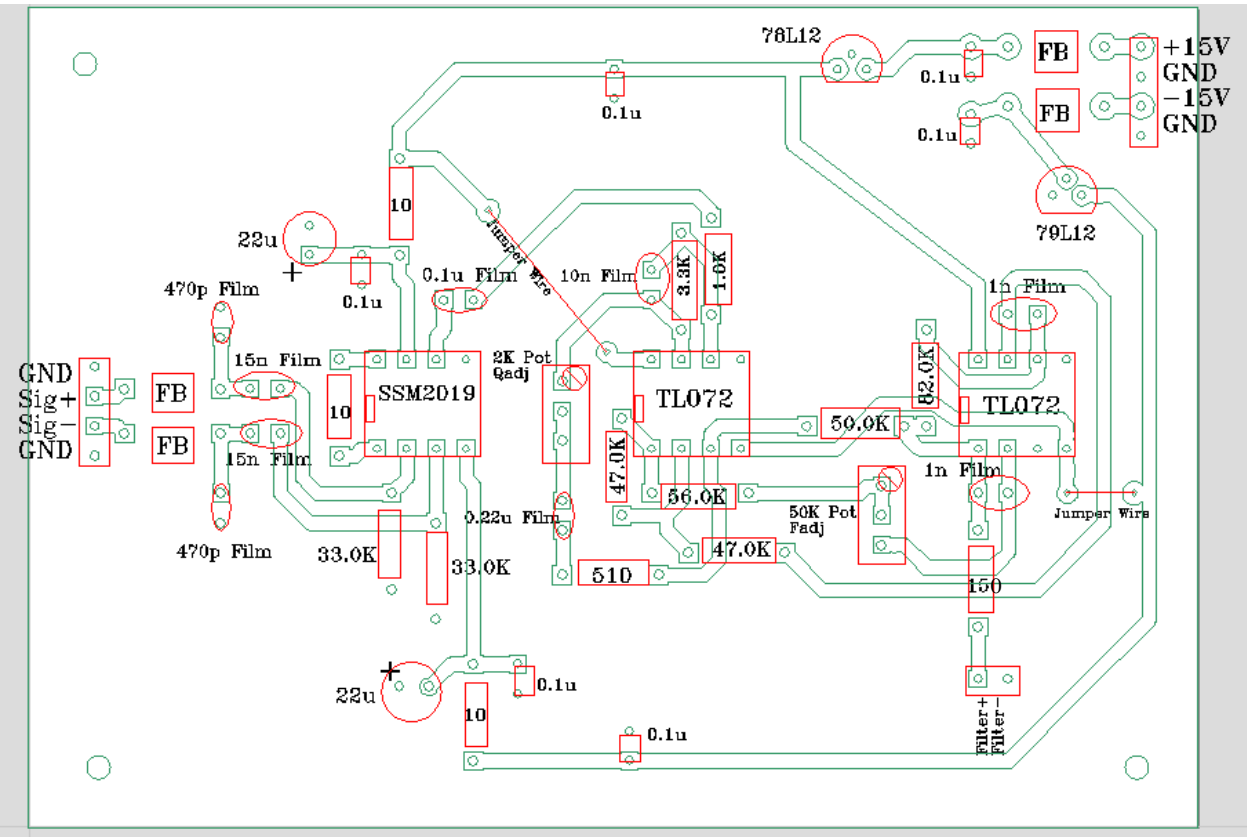
After 600 turns were wound, the wire was fed through the top drill hole. The winding pack was protected with a wrap of tape.



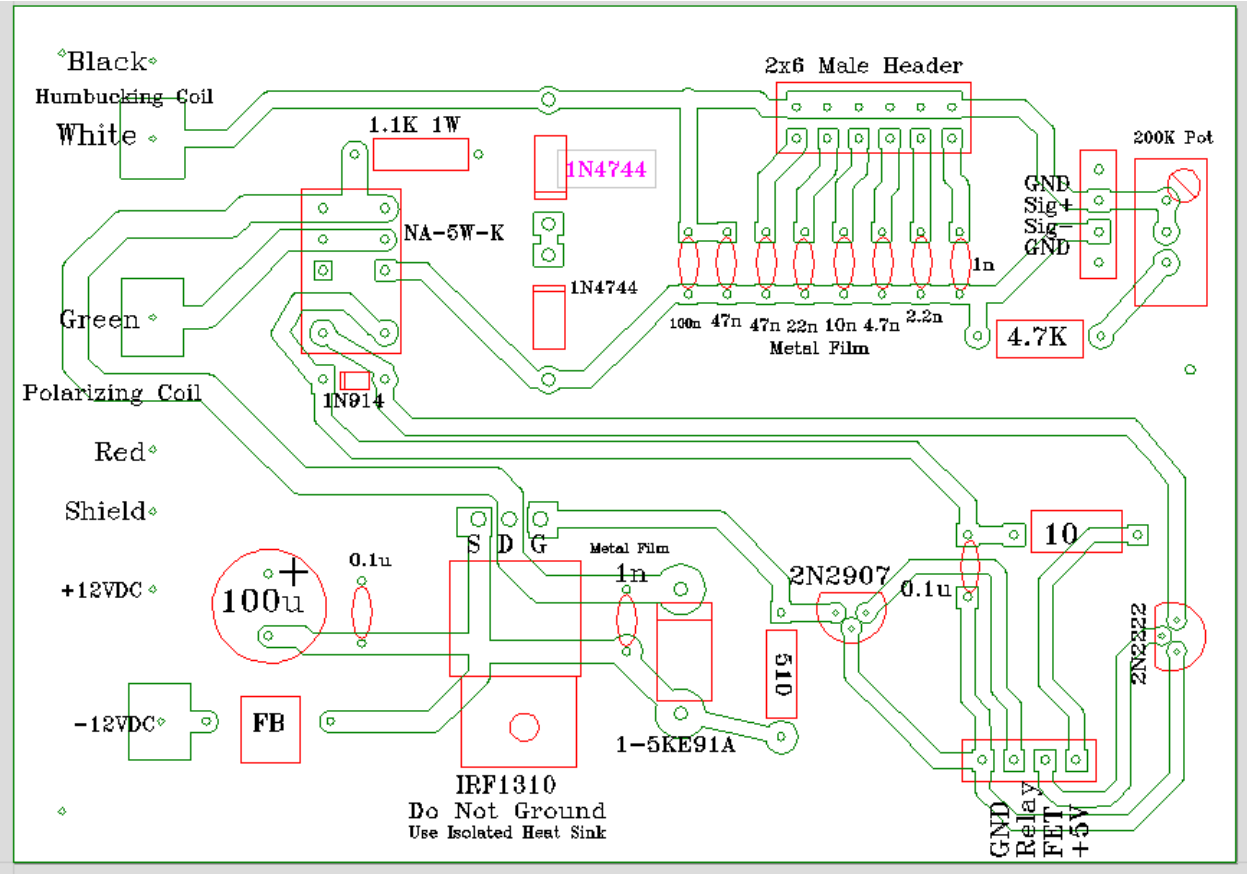
Both exit wires are secured to the form with a couple of wraps of tape



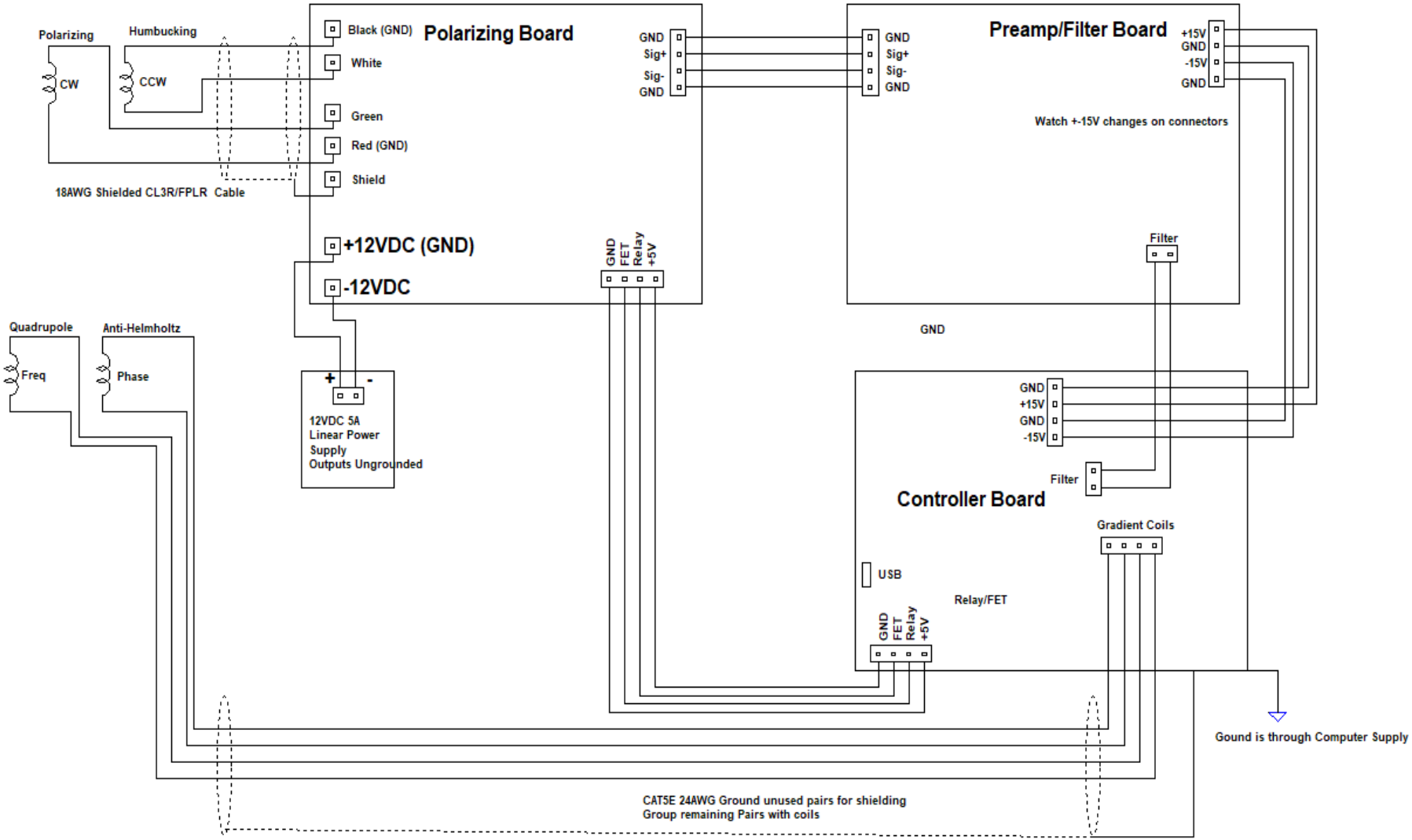
Component placement for the Controller board.



Component placement for the preamp/filter board. Use metal film resistors for the preamp and filter circuits. Supply resistors can be carbon film. Capacitors are metal film types. Polypropylene types are preferred over mylar film.



Component placement for the polarizing board.


Polarizing
Humbucking
CW
CCW
Black (GND)
White
Green
Red (GND)
Shield
18AWG Shielded CL3R/FPLR Cable
Polarizing Board
+12VDC (GND)
-12VDC
GND
Sig+
Sig-
GND
GND
FET
Relay
+5V
Preamp/Filter Board
GND
Sig+
Sig-
GND
+15V
GND
-15V
GND
Watch +-15V changes on connectors
Filter
GND
Quadrupole
Anti-Helmholtz
Freq
Phase
+
-
12VDC 5A
Linear Power
Supply
Outputs Ungrounded
GND
+15V
GND
-15V
Filter
Controller Board
Gradient Coils
USB
Relay/FET
GND
FET
Relay
+5V
Gound is through Computer Supply
CAT5E 24AWG Ground unused pairs for shielding
Group remaining Pairs with coils